System Design: Availability & Fault Tolerance

English study notes based on the supplied Hindi YouTube lecture transcript. The transcript contains missing and mistranscribed segments; unclear portions are summarized conservatively rather than presented as exact quotations.

1. What is Availability?

Availability means that a system or service is accessible and able to respond when users need it. The lecture uses the example of a Class 12 board-examination results website receiving a very large number of requests when results are published. If the website becomes unreachable at that moment, its availability is poor and the user experience suffers.

In practice, availability is commonly expressed as the percentage of time a service is operational: Availability = Uptime / (Uptime + Downtime) x 100%. This formula is an added study clarification, not a verbatim statement from the transcript.

2. Monolithic vs Distributed Systems

• Monolithic architecture: modules are deployed together. If the single host running the application fails, the entire application can become unavailable.

• Distributed architecture: components can run on separate servers. A failure in one server or component need not take down every other component, provided the system is designed to isolate failures.

• The lecture argues that distributed systems can achieve higher availability through replication and fault tolerance. Distribution alone does not guarantee availability; networking and dependencies can also fail.

3. Single Point of Failure (SPOF)

A single point of failure is a component whose failure can bring down the service. A monolith deployed on only one machine is the lecture's main example. To improve availability, avoid depending on a single machine, application instance, or critical infrastructure component.

4. Fault Tolerance and Availability

Fault tolerance is a system's ability to keep providing service despite failures. The lecture links greater fault tolerance with greater availability. For example, when one instance fails, another healthy instance can continue serving requests.

5. Replication

Replication means maintaining additional copies of a service or data so that a failure of one copy does not necessarily interrupt access. The lecture mentions replicating application servers and other infrastructure, including power sources.

• Application replicas: run multiple instances of the same service on different machines.

• Data replicas: maintain copies of data across nodes. Replication strategy and consistency matter, particularly for databases.

6. Redundancy vs Replication

The lecture distinguishes redundancy and replication, although the auto-generated transcript is especially unclear in this section.

• Redundancy: extra components or capacity kept available to take over when a component fails (for example, an additional server or power supply).

• Replication: copying and maintaining service instances or data across locations or nodes (for example, database replicas).

These concepts overlap: replication is one way to create redundancy. For applications, additional running instances can provide redundancy; for databases, replication involves data synchronization and recovery considerations.

7. Example Architecture

Users -> Load Balancer -> Application Instance A / Application Instance B -> Database with replica(s). If Instance A fails, a health-aware load balancer can route requests to Instance B. Database replicas may help with resilience, but automatic failover must be configured and tested.

8. Quick Interview Revision

• Q: What is availability? A: The ability of a service to remain accessible and operational when requested.

• Q: Why is a single-server monolith risky? A: A host failure can become a single point of failure for the entire application.

• Q: How can availability be improved? A: Use redundant instances, replication, failover, health checks, and remove single points of failure.

• Q: Is a distributed system always more available? A: No. It can improve fault isolation and redundancy, but it adds network and dependency failure modes.

• Q: What is the difference between replication and redundancy? A: Replication creates copies; redundancy means having spare or alternative capacity. Replication can provide redundancy.

Lecture Takeaway

The key message is that availability depends on fault tolerance. Replication and redundancy can reduce the impact of individual machine or component failures. The transcript emphasizes that distributed deployments offer more opportunities to build such resilience than a single-server deployment.